

Experiment: 4 To solve a Crypt-Arithmetic problem using Python.

Aim

To solve a Crypt-Arithmetic problem using Python.

Algorithm

1. Assign a unique digit to each letter.
2. Ensure no two letters have the same digit.
3. Convert the words into numbers.
4. Check whether the arithmetic equation is satisfied.
5. Display the solution if found.

Program

```
from itertools import permutations

letters = 'SENDMORY'

for p in permutations(range(10), 8):

    S,E,N,D,M,O,R,Y = p

    if S == 0 or M == 0:

        continue

    SEND = 1000*S + 100*E + 10*N + D

    MORE = 1000*M + 100*O + 10*R + E

    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:

        print("SEND =", SEND)

        print("MORE =", MORE)

        print("MONEY =", MONEY)

        break
```

```
exp 4 cript arthmatic.py - C:/Users/yadal/OneDrive/Documents/AI/exp 4 cript arthmatic.py (...
File Edit Format Run Options Window Help
from itertools import permutations

letters = 'SENDMORY'

for p in permutations(range(10), 8):
    S,E,N,D,M,O,R,Y = p

    if S == 0 or M == 0:
        continue

    SEND = 1000*S + 100*E + 10*N + D
    MORE = 1000*M + 100*O + 10*R + E
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        break
```

Output

```
>>|==== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 4 cript arthmatic.py ====
SEND = 9567
MORE = 1085
MONEY = 10652
>>|
```

Result

The Crypt-Arithmetic problem **SEND + MORE = MONEY** was successfully solved using Python, and the correct digit assignment was obtained.